WORKED SOLUTION

STEP III 2006, Question 11

Mechanics • Collisions in Connected Systems

STEP III 2006, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q11

The key idea

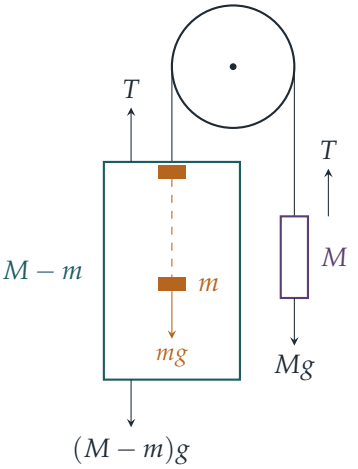
Once the tile lets go, the lift is lighter than the counterweight, so the lift starts to rise while the tile falls: the tile has to cover the height h *relative to the lift*. For the impact, the lift and counterweight act as a single body of mass $2M - m$ joined to the tile through the floor, and the given condition on the cable's impulses is exactly what lets us write a momentum-type conservation law for that body and the tile. That "momentum" turns out to be zero just before impact, which makes the energy bookkeeping very clean.

Solution

We model the lift, counterweight and tile as particles (for vertical motion), the cable as light and inextensible, and the pulley as smooth. Before the tile detaches the system is at rest, so the lift *including* the tile balances the counterweight: the lift has mass M with the tile, and $M - m$ without it.

The fall

Take upwards as positive for the lift and the tile. Since the cable is inextensible, when the lift moves up with speed u and acceleration a, the counterweight moves down with the same speed and acceleration. Let T be the tension.



Newton's second law for the lift (upwards) and the counterweight (downwards) gives

$$(M-m)a = T-(M-m)g, \qquad Ma = Mg - T.$$

Adding, $(2M-m)a = mg$, so

$$a = \frac{mg}{2M-m}.$$

The tile is in free fall with acceleration g downwards (the guides are smooth and nothing else touches it). Relative to the lift, the tile therefore accelerates downwards at

$$g + a = g + \frac{mg}{2M-m} = \frac{2Mg}{2M-m},$$

starting from rest relative to the lift. It meets the floor after a time t with $\frac{1}{2}(g+a)t^2 = h$, so

$$t^2 = \frac{(2M-m)h}{Mg}.$$

At that moment the tile's speed is

$$v = gt = \sqrt{g^2 \cdot \frac{(2M-m)h}{Mg}} = \sqrt{\frac{(2M-m)gh}{M}},$$

as required. The lift (and counterweight) then have speed

$$U = at = \frac{m}{2M-m}\,gt = \frac{mv}{2M-m}.$$

In this time the lift rises $\frac{1}{2}at^2 = \dfrac{mh}{2M} < h$, so the condition on the distance to the pulley guarantees the lift doesn't reach the pulley first.

The impact

Let J be the impulse of the floor on the tile (upwards), and let I be the impulse of the cable's tension, which by the given condition has the same magnitude on the lift and on the counterweight (it acts upwards on each). Gravity gives no impulse during the instantaneous impact. Let the tile's velocity just after be w (upwards), and let the lift's velocity just after be U' (upwards), so the counterweight's downward velocity is also U'. Then:

$$\begin{aligned} \text{tile:} \quad & m\big(w-(-v)\big) = J, \\ \text{lift:} \quad & (M-m)(U'-U) = I - J, \\ \text{counterweight (downwards):} \quad & M(U'-U) = -I. \end{aligned}$$

Adding all three eliminates J and I:

$$mw + (2M-m)U' = -mv + (2M-m)U.$$

Ask yourself Why is it legitimate to add the counterweight's equation, which uses "downwards" as positive, to the others?

Because of the cable, "counterweight down" and "lift up" are the same coordinate. What we have shown is that the combination $mw + (2M-m)U'$ (tile's upward momentum plus the momentum of lift and counterweight measured along the cable) is unchanged by the impact. This is where the equal-impulse condition is used: without it, I would not cancel.

Now the key observation: since $U = \dfrac{mv}{2M-m}$, the right-hand side is $-mv + mv = 0$. So

$$mw + (2M-m)U' = 0 = -mv + (2M-m)U.$$

Newton's experimental law at the floor says that the speed of separation is e times the speed of approach:

$$w - U' = e\big(U-(-v)\big) = e(U+v).$$

Both the before and after velocities satisfy the same "zero momentum" equation, so each of the pairs (w, U') and $(-v, U)$ is a multiple of $(2M-m,\ -m)$. The restitution equation then fixes the multiple: $w = ev$ and $U' = -eU$. (Check: $mev - (2M-m)eU = e(mv - mv) = 0$, and $w - U' = e(v+U)$.)

The energy loss

Just before impact, the kinetic energy is

$$\tfrac{1}{2}mv^2 + \tfrac{1}{2}(2M-m)U^2 = \tfrac{1}{2}mv^2\left(1 + \frac{m}{2M-m}\right) = \frac{Mmv^2}{2M-m} = mgh,$$

using $U = \frac{mv}{2M-m}$ and then $v^2 = \frac{(2M-m)gh}{M}$. Just after, every velocity has been multiplied by $\pm e$, so the kinetic energy is $e^2\,mgh$. The impact is instantaneous, so the potential energy doesn't change during it, and the loss of energy of the system is

$$mgh - e^2 mgh = mgh(1-e^2),$$

as required.

Comment

The answer does not depend on M at all: it is exactly the loss we would get if the tile fell a height h inside a *fixed* lift and bounced with coefficient e. That makes sense. Just before impact the system's kinetic energy is mgh, which is simply the potential energy released (the tile and the lift close up a gap of h, and the lift's rise is exactly balanced by the counterweight's fall). And because the "momentum" along the cable is zero, the impact behaves like a collision viewed in the zero-momentum frame, where a fraction $1 - e^2$ of the kinetic energy is always lost. In particular, if $e = 0$ all the kinetic energy is lost and the whole system comes to rest; if $e = 1$ nothing is lost.

Checks

- Take $M = 5$, $m = 2$, $h = 3$, $g = 10$, $e = \frac{1}{3}$. Then $v^2 = 48$ and $U^2 = 3$, so the kinetic energy before impact is $\frac{1}{2} \cdot 2 \cdot 48 + \frac{1}{2} \cdot 8 \cdot 3 = 60 = mgh$, and the loss is $60 \cdot \frac{8}{9} = \frac{160}{3}$.
- Stepping the motion numerically (tile in free fall, lift accelerating up at $\frac{mg}{2M-m}$) for 200 random choices of M, m, h, e reproduces the impact speed v, the lift speed U and the loss $mgh(1 - e^2)$.
- As $m \to 0$ the lift hardly moves and $v \to \sqrt{2gh}$, the free-fall speed, as it should.
- The tension impulse $I = M(U - U') = M(1 + e)U$ is positive, so the cable stays taut during the impact.

Common slips

- Treating the lift as having mass M after the tile has fallen. Then nothing would accelerate and the tile's speed would be $\sqrt{2gh}$.
- Forgetting that the floor rises to meet the tile, so the tile falls less than h relative to the ground.
- Applying conservation of momentum to the tile and lift alone, forgetting the counterweight, which is linked to the lift through the cable.

Beyond STEP

The quantity conserved in the impact is not ordinary momentum but the momentum conjugate to the single coordinate describing the cable's position: a first taste of generalised coordinates and Lagrangian mechanics. The fact that a collision with zero total momentum loses exactly the fraction $1 - e^2$ of the kinetic energy is the centre-of-mass frame idea that runs through all of collision theory. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both momentum in the centre-of-mass frame and systems of connected bodies carefully.

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